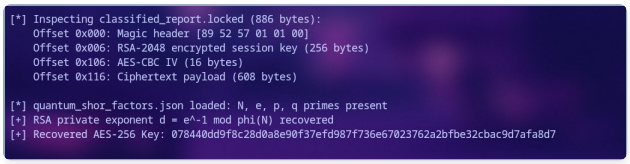
NullVector Ransomware

Cryptography



I examined the provided files `classified_report.locked` (886 bytes) and `quantum_shor_factors.json`. The container started with a 6-byte magic header, followed by a 256-byte RSA wrapped key block and an AES-256-CBC ciphertext.

The JSON file provided the RSA-2048 modulus N, public exponent e, and prime factors p and q, allowing immediate calculation of private exponent d.

After unwrapping the AES key and decrypting the CBC layer, I analyzed the 597-byte stream and found repeated 46-byte section dividers. Inverting the 5-bit circular right shift and affine linear mask recovered the original intel report.

Finally, I tested the recovered flag against the executable / service:

`python3 NullVector_Ransomware.py`

Classified report decrypted. Header verification passed.

Flag Captured

`SecLeaf{e4d90b218a754f1c93206e58b1f4a7c2}`

BY:JOKER_53